Slopes and local change

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Image-pattern Network Investigation

Prepare the purpose, prerequisites and materials

> Prepare Slopes and local change: check the target, prerequisites, resources and adaptable teaching schedule.

Teach the learning cycle

Primary target: Calculate an average rate of change over a stated interval

Session · Sequence and minutes · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Sequence and minutes:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Use Your mission, Notice the evidence, See how it works, Words that unlock the idea and Follow a worked example; record recall in the workbook.

> **Session:** Session 2 · Practise · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
>
> **Resources:** Use Find and repair a mistake, Practise together and Build your understanding; use the matching workbook W2 and Individual reasoning check.

> **Session:** Session 3 · Investigate and transfer · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Use Run the investigation, Connect your project and Try a new case independently. Allow extra time for the full task set. When code is supplied, use the examples in See how it works or Follow a worked example and the browser lab in Supported practice, repair and portfolio.

This is a starting allocation of 135 minutes, not a validated duration. Preserve all named topics. Schedule extra time for connected examples, the full investigation, independent cases and project building.

Repeated copies of a case share one recorded attempt; they are not new **evidence**.

Retrieval answer guidance

Loss functions

Without opening the old lesson, explain one key idea from Loss functions. Give an example and a mistake that the idea helps you avoid.

Look for: Calculate per-example squared error and mean squared error

Reference example, not the only acceptable wording: Teacher explanation: Residuals are -1 and 2, squares 1 and 4, MSE 2.5.

Forward passes by hand

Without opening the old lesson, explain one key idea from Forward passes by hand. Give an example and a mistake that the idea helps you avoid.

Look for: Record scores and activations in dependency order

Reference example, not the only acceptable wording: Teacher explanation: z = [-3, 3], h = [0, 3], y = -5.

Bias terms and decision boundaries

Without opening the old lesson, explain one key idea from Bias terms and decision boundaries. Give an example and a mistake that the idea helps you avoid.

Look for: Calculate a weighted score with a bias term

Reference example, not the only acceptable wording: Teacher explanation: Weighted sum 10 - 4 = 6; bias -3; score 3; **prediction** 1. Add the bias once after summing.

Introduce rate of change through finite differences, with no assumed calculus fluency. For the compact alternative schedule, use two 45-minute sessions, a calculator and a labelled parameter-versus-loss sketch.

Reserve the word local for a stated centre and nearby values.

Outcomes

Calculate an average rate of change over a stated interval

Estimate a local **slope** with nearby loss values

Explain the limits of a local direction clue

Preparation

Printed lesson and workbook; paper and pencil

Calculator and optional local Python 3 editor; no packages or AI accounts needed

Use the fictional examples supplied; prepare the fresh retry before class

Prerequisite answer

The residual is 2 and the squared error is 4.

End goal for the pictorial case

By the end, I can explain **slope**s and **local change** using a traced example and a stated limitation.

This added worked case illustrates the week’s main idea. Retain all original and connected topics; schedule its practice within the teaching cycle.

Explain and model the mechanism

> Model Slopes and local change using the supplied case, labelled mechanism and reasoning. Make the scope of each conclusion explicit.

Let a one-example model predict w and let its target be 2. Its squared loss is L(w) = (w - 2) squared. Here w is the parameter we vary, not a new data example. L is a nonnegative function of that parameter.

From w = 0 to w = 1, the loss changes by -3 while w changes by 1. The average **slope** is -3. From w = 1 to w = 2 it is -1. The curve does not have one fixed **slope** everywhere.

Near w = 1, use two symmetric nearby values: 0.9 and 1.1. Their losses are 1.21 and 0.81. Change in loss divided by change in w gives (0.81 - 1.21) / (1.1 - 0.9) = -2.

This symmetric estimate describes change near the centre w = 1.

For this quadratic, the symmetric estimate equals the exact **slope** 2(w - 2) in exact arithmetic. Numerical estimates for other functions are generally approximate.

Smaller steps can reduce some approximation errors, but extremely small steps can suffer from computer rounding.

Term	Meaning
Slope	Change in output divided by change in input over a stated interval
Local change	Change considered near a particular input or parameter value
Step size	The amount by which an input or parameter is changed
Stationary point	A point where a differentiable function has zero local slope

Compare slope clues on the same loss curve

CURRENT W	LOCAL SLOPE	NEARBY DOWNHILL DIRECTION
1	-2	Increase w a little
2	0	No first order direction
3	2	Decrease w a little

These are local clues for L(w) = (w - 2) squared. At w = 2 the displayed function has its minimum; zero slope does not establish a minimum for every possible function.

Guided: L(2.9) = 0.81 and L(3.1) = 1.21, so **slope** is 0.4 / 0.2 = 2. A small decrease in w lowers this loss. Exit: at w = 1 **slope** is negative, but increasing w to 4 raises loss from 1 to 4.

Model the reasoning aloud

Use the complete worked example below before asking learners to complete W2. Reveal a small part, invite a **prediction**, then explain the deciding step.

These teacher notes contain solutions; do not reveal the independent case answer during modelling.

Compare slope clues on the same loss curve

CURRENT W	LOCAL SLOPE	NEARBY DOWNHILL DIRECTION
1	-2	Increase w a little
2	0	No first order direction
3	2	Decrease w a little

These are local clues for L(w) = (w - 2) squared. At w = 2 the displayed function has its minimum; zero slope does not establish a minimum for every possible function.

At w = 3, use 2.9 and 3.1. Calculate both losses and the symmetric slope. Which small change should lower loss here?

Thinking move	Teacher prompt
Plan	What is given? What is the question? Which procedure fits these facts?
Do	Point to each input and intermediate step in the worked trace. Name the rule that allows the next step.
Check	Does the output answer the question? Which assumption or limitation prevents a larger claim?

Use the diagram and verbal explanation together. Ask learners to match a sentence to a specific visual step; do not assign learners fixed visual or auditory learning styles.

A pictorial worked case: Slopes and local change

Use a local **slope** to propose a small move rather than claiming every move helps.

The facts we will use

Supplied facts

Toy loss $J(a)=(a-2)^2$. At $a = 0$, use symmetric step 0.5.

Supplied facts

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

AI ACADEMY / AI-5/9

Slopes and local change

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.

1. Evaluate nearby points

J(0.5)=2.25 and J(-0.5)=6.25.

↓

2. Estimate slope

Slope=(2.25-6.25)/(0.5-(-0.5))=-4.

↓

3. Check small move

Move to a = 0.25: loss 3.0625, below starting loss 4.

CHANGE ONE THING

Move from 0 all the way to5.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Evaluate nearby points	J(0.5)=2.25 and J(-0.5)=6.25.
2. Estimate slope	Slope=(2.25-6.25)/(0.5-(-0.5))=-4.
3. Check small move	Move to a = 0.25: loss 3.0625, below starting loss 4.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

Why the result follows

A **slope** describes how the loss changes near the current parameter. The symmetric calculation compares equal small moves on either side. Its negative result suggests a small increase may help at zero.

It does not promise every increase helps: jumping to five passes the optimum at two and produces a larger loss. Check the new loss after each proposed step.

What this example does not establish

Keep the stated limit

A finite difference approximates **local change**; numerical precision and nonsmooth points need care in more complex objectives.

Change one thing in the pictured case

Change one thing

Change: Move from 0 all the way to5.

Predicted result

Prediction: Loss becomes 9, exceeding the starting 4.

Why the result follows

Reason: A negative local **slope** suggests increasing a locally, not that every positive jump decreases loss.

Ask while showing the picture

Ask while showing the picture · 1

Cover the result. Ask learners to name the **input**s and predict the next step. Uncover one step, connect the icon to the actual procedure, and explain its **evidence**. Then change the supplied condition and compare.

The picture is a representation; it does not document execution of a real AI model.

Demonstrate and guide practice

> Demonstrate Slopes and local change, reduce support deliberately and check individual reasoning before discussing answers.

Time or stage	Teacher action
Session 1 start to 7 minutes	Retrieve squared loss and distinguish parameter w from an example input.
7 to 18 minutes	Build the five-row table and sketch the curve. Calculate slopes across two different intervals.
18 to 32 minutes	Complete W1 and the worked symmetric estimate near 1. Emphasise denominator 2 times step.
32 to 45 minutes	Run or trace the optional code, complete W2 and the guided case.
Session 2 start to 8 minutes	Compare small changes with jumps across the minimum using W3.
8 to 20 minutes	Use F(w) to challenge zero-slope claims and complete W5.
20 to 35 minutes	Collect W4 independently, including interval arithmetic and the oversized-step counterexample.
35 to 45 minutes	Use the exit response and a fresh retry to check direction, scale and stated limits.

Use See how it works and Follow a worked example. Collect a **prediction** before revealing its result; ask the learner to point to each intermediate fact.

Provide an axis label and the fraction change in loss over change in parameter.

Extension: algebraically expand the symmetric difference for (w - 2) squared to recover 2(w - 2), noting exact arithmetic versus floating-point computation.

Reduce help deliberately

Completion practice

Completion task: At centre w = 0, use step 0.5. Calculate L(-0.5), L(0.5) and the symmetric **slope**.

Reasoning to listen for

Teacher check: Teacher explanation: Losses are 6.25 and 2.25. Denominator is 1, giving **slope** -4.

Start by identifying the given **input**s together. Leave the intermediate steps blank. If needed, model one step, then withdraw that help on the next step. Move to a full trace only when the partial trace is understood.

For partners, alternate explainer and checker; collect a response from every learner before discussion.

Individual reasoning check

Use the worked case for **Slope**s and **local change**. Proposed explanation: “Divide by the step only”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

Expected repair

Expected correction: A symmetric comparison spans from centre minus step to centre plus step, a width of twice the step.

Collect explanations, not only yes/no votes. If a misunderstanding is common, re-model the relevant step for the class.

If it affects a few learners, give a short targeted group explanation while others solve a more demanding variation.

Pictorial practice question

Which calculation disproves “any increase helps”? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Reasoned teacher answer

$J(5)=(5-2)^2=9>J(0)=4$. The local direction is useful only with an appropriately sized step and verification. Changed-case check: Loss becomes 9, exceeding the starting 4.

A negative local **slope** suggests increasing a locally, not that every positive jump decreases loss.

Changed-case reasoning: Loss becomes 9, exceeding the starting 4. A negative local **slope** suggests increasing a locally, not that every positive jump decreases loss.

Collect each learner's explanation before discussion. Use the first missing connection between **input**, deciding step and result to choose support. This question is worked-case practice, not fresh mastery **evidence**.

Run the investigation and diagnose misconceptions

> Run the Slopes and local change investigation and use the first unsupported step to choose a teaching response.

Divide by the step only: A symmetric comparison spans from centre minus step to centre plus step, a width of twice the step.

Negative loss **slope** means negative loss: The **slope** describes change; squared loss itself remains nonnegative.

Zero **slope** always means finished: **Stationary point**s need additional **evidence**; some are not minima.

Use Run the investigation for the **evidence** ledger. Add a boundary or missing-**input** case and record the expected response before checking. More successful repeats of one case do not establish wider coverage.

Match the support to the first error

What the response shows	Next teaching action
Missing or misread facts	Read the case aloud, mark supplied versus unknown facts, and let the learner restate it.

What the response shows	Next teaching action
Wrong procedure	Return to the deciding step in the worked case. Contrast it with the proposed incorrect explanation.
Arithmetic or implementation error	Trace intermediate values or lines; preserve the first mismatch and retry that step.
Correct result without a reason	Ask for a trace and a changed case before marking independent understanding.
An unsupported wider claim	Ask which supplied evidence supports the claim and which further observation would be needed.

Offer an oral explanation, drawing, enlarged visual or fewer examples when that removes a reading/writing barrier. Keep the same reasoning goal.

Stretch secure learners by changing a boundary or assumption and asking them to defend the effect, not simply by assigning more of the same questions.

Assess and explain the answers

> Use the Slopes and local change questions, teacher keys and criteria after the relevant attempts. Previously discussed cases are practice; protect the separately selected independent task.

Guided and exit questions

Guided: L(2.9) = 0.81 and L(3.1) = 1.21, so **slope** is 0.4 / 0.2 = 2. A small decrease in w lowers this loss. Exit: at w = 1 **slope** is negative, but increasing w to 4 raises loss from 1 to 4.

W1 Measure an interval slope

Workbook question · W1

For L(w) = (w - 2) squared, calculate the average **slope** from w = 0 to w = 2. Compare it with the local **slope** at w = 2.

Teacher reasoning · W1

Teacher explanation: Average **slope** is (0 - 4) / 2 = -2. The local **slope** at 2 is zero. They describe different locations or intervals.

W2 Check a symmetric estimate

Workbook question · W2

At centre w = 0, use step 0.5. Calculate L(-0.5), L(0.5) and the symmetric **slope**.

Teacher reasoning · W2

Teacher explanation: Losses are 6.25 and 2.25. Denominator is 1, giving **slope** -4.

W3 Challenge a large step

Workbook question · W3

Starting at w = 3, compare loss after moving to 2.9 and after moving to -1. Both moves decrease w.

Teacher reasoning · W3

Teacher explanation: Initial loss is 1; at 2.9 it is 0.81; at -1 it is 9. The small move helps, the large move overshoots.

W4 Estimate a new curve independently

Workbook question · W4

Let J(a) = (a + 1) squared. At a = 1 use step 0.25 to find the symmetric **slope**. Compare J(1) with J(0.9), then explain why moving to -4 does not follow safely from the **slope** clue.

Teacher reasoning · W4

Teacher explanation: J(0.75) = 3.0625; J(1.25) = 5.0625; **slope** is 2 / 0.5 = 4. J(1) = 4 and J(0.9) = 3.61, so a small decrease helps. J(-4) = 9: a large decrease overshoots.

W5 Limit zero slope evidence

Workbook question · W5

For F(w) = (w squared - 1) squared, compare F(0) and F(0.1). Does zero local **slope** at zero prove a minimum?

Teacher reasoning · W5

Teacher explanation: F(0) = 1 and F(0.1) = 0.9801. Nearby loss is lower, so zero **slope** alone does not prove a minimum.

Independent assessment and fresh retry

Teacher key · Independent assessment and fresh retry

W4 requires losses 3.0625 and 5.0625, **slope** 4, and the finite-step limitation. Fresh retry: K(c) = (c - 3) squared at c = 2, step 0.5, uses losses 2.25 and 0.25 and **slope** -2. Moving to 2.1 gives loss 0.81 rather than 1.

Assess independent transfer

Independent question · withhold the key

Selected case: Let J(a) = (a + 1) squared. At a = 1 use step 0.25 to find the symmetric **slope**. Compare J(1) with J(0.9), then explain why moving to -4 does not follow safely from the **slope** clue.

Selected case answer

Teacher key · Selected case answer

Teacher explanation: J(0.75) = 3.0625; J(1.25) = 5.0625; **slope** is 2 / 0.5 = 4. J(1) = 4 and J(0.9) = 3.61, so a small decrease helps. J(-4) = 9: a large decrease overshoots.

Dimension	Evidence required
Explanation	Names the applicable idea or procedure and ties it to the supplied facts.
Trace and result	Preserves the given inputs and shows the relevant intermediate steps; distinguishes a prediction from an observed execution.
Limits	States the assumptions, missing evidence or conditions under which the result would change.

Teacher key · Selected case answer

Record each dimension as not yet, with support, or independent. Accept a valid alternative procedure with a defensible trace.

Do not penalise an honestly recorded model failure if the learner correctly explains its **evidence** and limits.

Feedback that the learner uses

Name the earliest unsupported step and give one action to repair it. Have the learner make that repair now and explain why. Then assess the relevant skill on a fresh, fully specified case.

Confirm it was not previously practised with feedback; solve and check the new case before assigning it. A follow-up design prompt is a starting brief, not automatically an equivalent test.

Connect the project, support and next lesson

> Connect Slopes and local change to Image-pattern Network Investigation, adapt support and plan a later changed-case check.

Add a labelled **slope**s and **local change evidence** sheet to Image-pattern Network Investigation. Show how you can calculate an average rate of change over a stated interval.

Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

Provide an axis label and the fraction change in loss over change in parameter.

Extension: algebraically expand the symmetric difference for (w - 2) squared to recover 2(w - 2), noting exact arithmetic versus floating-point computation.

Week 10 extends local **slope**s to several parameters.

Retain the project ledger and assess a changed case after support. Check the relevant prerequisite before carrying this component forward.

Plan the delayed return

Suggested return points in this level: ai-5/10, ai-5/12, ai-5/15. Use actual classroom dates, adjusting for holidays and gaps. One, three and six teaching weeks are scheduling choices, not a claimed optimal spacing.

At a delayed check, ask for the idea from memory and a changed application. Compare with the earlier independent response and record support used.

For a new level, revisit the previous capstone before checking the new prerequisites.

Record	Teacher evidence
Learner code and date	
Concept and case used; prior exposure	
Explanation / trace / limits	
Support and first repair	
Changed case and delayed outcome	
Next teaching action	

Use the **evidence** to choose reteaching or the next step. A completed page, confidence rating or attractive project does not by itself demonstrate understanding.

Project evidence from this case

For your Image-pattern Network Investigation, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Prepare, teach and assess

> Use this preparation summary alongside the detailed Slopes and local change lesson; it is a checklist rather than a replacement for the full sequence.

Teaching target

Teaching target: Calculate an average rate of change over a stated interval

Prepare the labelled worked-case picture, the matching workbook, fictional **input** cards and a place for individual responses. Check a local Python 3 environment before class; have a paper-trace version ready.

Session · Plan · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Plan:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Use Your mission, Notice the evidence, See how it works, Words that unlock the idea and Follow a worked example; record recall in the workbook.

> **Session:** Session 2 · Practise · 45 min
>
> **Plan:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
>
> **Resources:** Use Find and repair a mistake, Practise together and Build your understanding; use the matching workbook W2 and Individual reasoning check.

> **Session:** Session 3 · Investigate and transfer · 45 min
>
> **Plan:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.

> **Resources:** Use Run the investigation, Connect your project and Try a new case independently. Allow extra time for the full task set. When code is supplied, use the examples in See how it works or Follow a worked example and the browser lab in Supported practice, repair and portfolio.

Before class, solve the supported practice: At centre w = 0, use step 0.5. Calculate L(-0.5), L(0.5) and the symmetric **slope**.

Supported practice answer

Reasoned practice answer: Teacher explanation: Losses are 6.25 and 2.25. Denominator is 1, giving **slope** -4.

Check every learner before discussion: Use the worked case for **Slope**s and **local change**. Proposed explanation: “Divide by the step only”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Repair the misconception

Misconception repair: A symmetric comparison spans from centre minus step to centre plus step, a width of twice the step.

Support without giving away the answer

Support: read the exact case aloud, enlarge the labelled picture, supply **input** cards and model only the first deciding step. Accept a spoken explanation or drawing; retain the reasoning goal.

Extend the reasoning

Extension: Move from 0 all the way to5. Require a **prediction** and a reason before comparing. Expected result: Loss becomes 9, exceeding the starting 4.

Reason: A negative local **slope** suggests increasing a locally, not that every positive jump decreases loss.

Protect the independent attempt

Independent assessment: withhold the reserved answer until the learner has attempted the new case. Record support separately from correctness.

Criterion	Not yet	With support	Independent
Explanation	Names an answer without a deciding fact	Uses a hint to link fact and decision	Explains the deciding fact in own words
Trace or test	Cannot connect input and result	Completes steps with prompts	Traces or tests a changed case accurately
Limits and revision	Claims more than evidence supports	Names a limit after a prompt	States a defensible limit and useful next test

Project checkpoint

Project checkpoint: Make a transparent first version. Keep a trace of how one **input** becomes an **output**; explain the rule or learned component.

Use the weekly teacher answer for topic-specific reasoning. A saved progress tick is an activity record; judge mastery from the learner's explanation and **evidence**.

Use feedback, labs and repair paths

Use feedback, earlier review and portfolios purposefully. Apply the coding routine only when

> this lesson has an executable lab.

Use the two formative questions as supported practice, not as independent mastery scores. Discuss the deciding **evidence** rather than accepting a guessed selection.

The feedback comes from the reviewed public worked case. Do not reveal the separately reserved independent teacher answer until the learner has attempted that case.

Ask students to predict before running code. Preserve errors and actual **output**s; collect a changed test and an explanation. The branch-boundary and indexing lessons deliberately include failure cases.

Use the earlier worked-explanation links to address a specific missing building block, then return to the current question. Adapt this suggestion after observing the learner; it is not an automatic diagnosis.

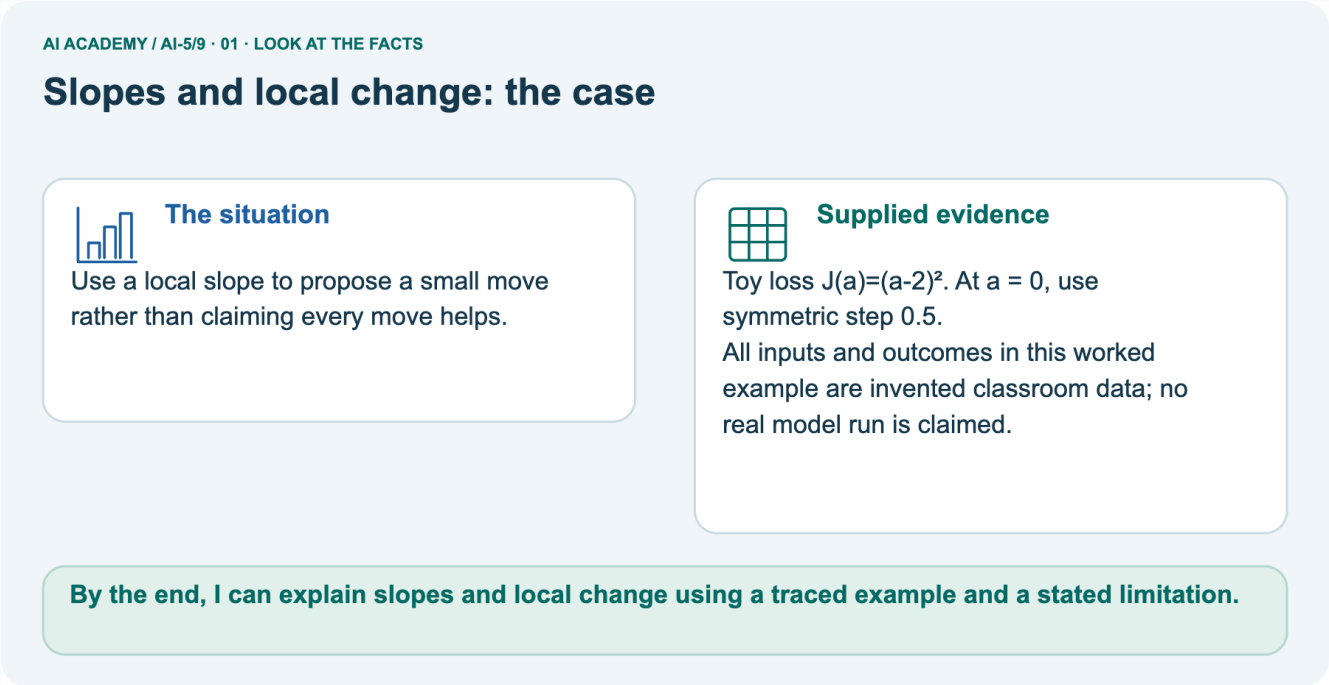
Collect the completed-work portfolio as **evidence**. Record support separately from correctness, and assess explanation, trace/test and limitations. Progress ticks and quiz selections alone do not establish mastery.

Visual concept journey

> Show one Slopes and local change graphic at a time, request a prediction and discuss its trace before the independent assessment.

Use one picture at a time. Hide a worked result before asking for a **prediction**, then reveal it for feedback. These are supported teaching cases; retain the separate independent assessment for a fresh first attempt.

01. Slopes and local change: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

Use a local **slope** to propose a small move rather than claiming every move helps. Supplied facts: Toy loss J(a)=(a-2)². At a = 0, use symmetric step 0.5.

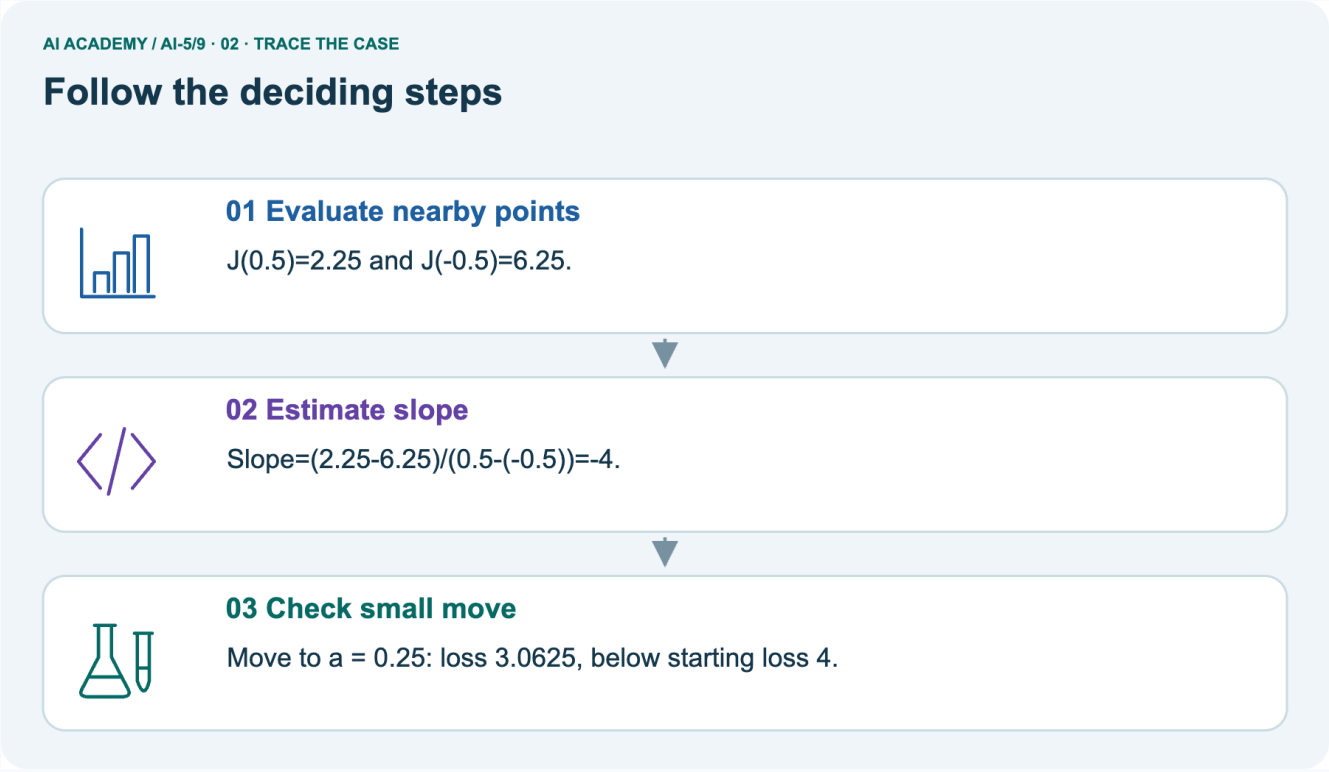
All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

Point to the given fact that matters most. Explain why it matters.

Discuss while showing the picture

Teacher prompt: Point to the given fact that matters most. Explain why it matters. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

02. Follow the deciding steps



Follow the labelled picture with the matching explanation. Open the vector diagram

Evaluate nearby points: J(0.5)=2.25 and J(-0.5)=6.25. → Estimate **slope**: **Slope**=(2.25-6.25)/(0.5-(-0.5))=-4. → Check small move: Move to a = 0.25: loss 3.0625, below starting loss 4.

Point to each numbered stage and explain which recorded **input** it uses.

Discuss while showing the picture

Teacher prompt: Point to each numbered stage and explain which recorded **input** it uses. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

03. Compare slope clues on the same loss curve

AI ACADEMY / AI-5/9 · 03 · INSPECT THE LESSON MODEL

Compare slope clues on the same loss curve

Current w	Local slope	Nearby downhill direction
1	-2	Increase w a little
2	0	No first order direction
3	2	Decrease w a little

These are local clues for L(w) = (w - 2) squared. At w = 2 the displayed function has its minimum; zero slope does not establish a minimum for every possible function.

Follow the labelled picture with the matching explanation. Open the vector diagram

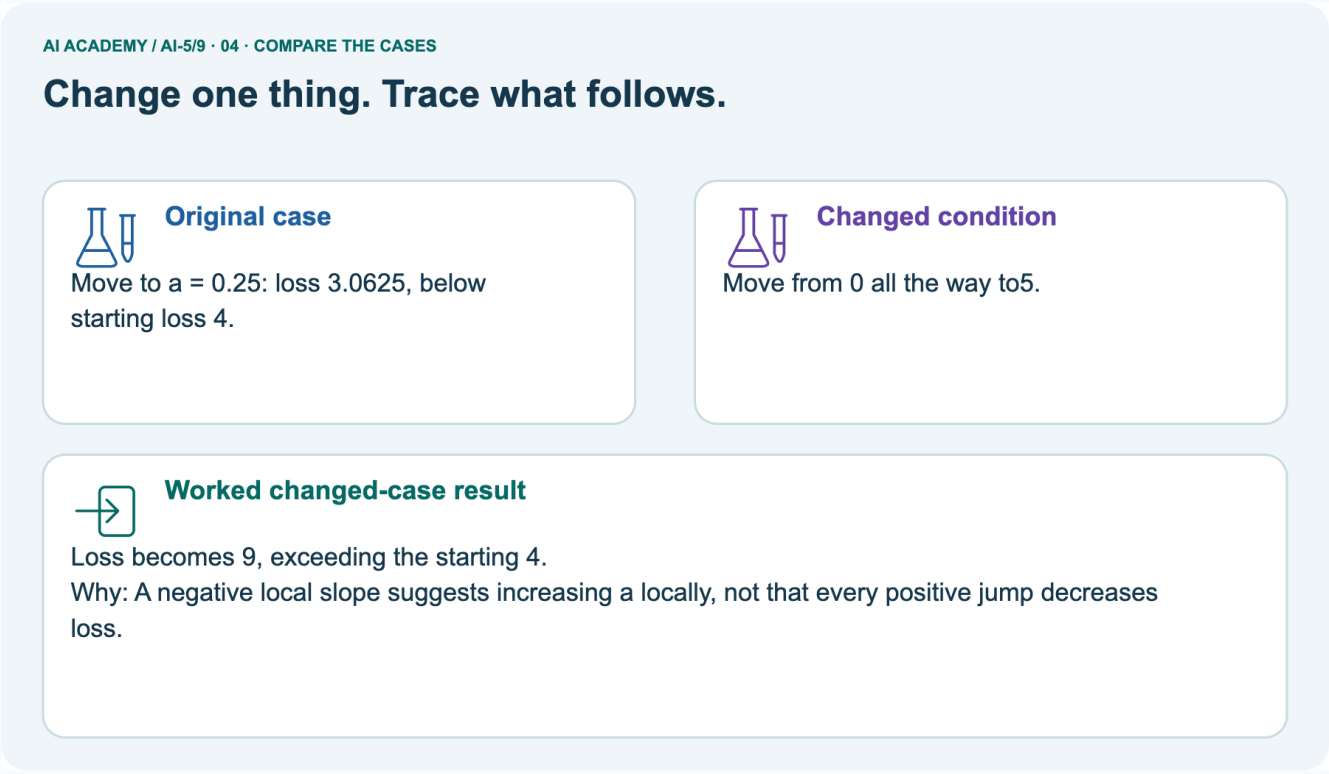
Compare **slope** clues on the same loss curve. 1 | -2 | Increase w a little. 2 | 0 | No first order direction. 3 | 2 | Decrease w a little

Read the labels and values. Explain the relationship the model shows.

Discuss while showing the picture

Teacher prompt: Read the labels and values. Explain the relationship the model shows. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Move to a = 0.25: loss 3.0625, below starting loss 4. Changed condition: Move from 0 all the way to5. Expected result: Loss becomes 9, exceeding the starting 4.

Reason: A negative local **slope** suggests increasing a locally, not that every positive jump decreases loss.

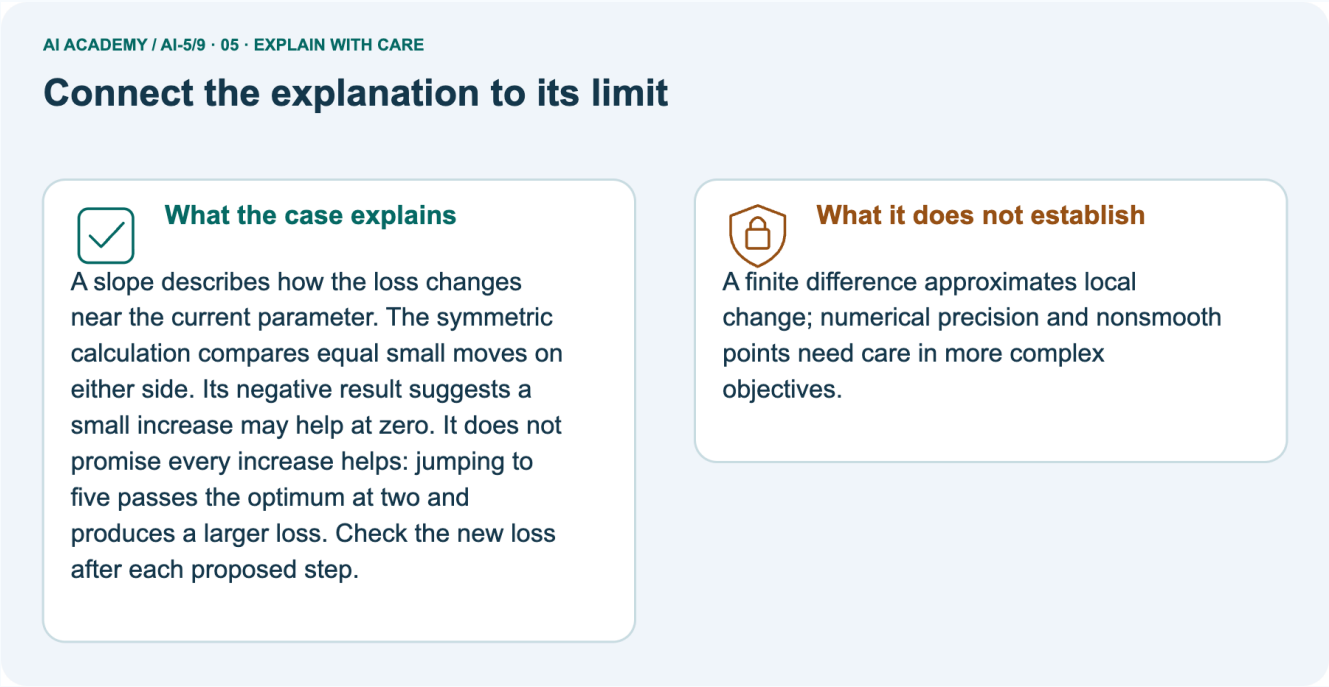
Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

Discuss while showing the picture

Teacher prompt: Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude. Ask learners to describe one label or arrow in ordinary words.

Use the adjacent transcript if the image is difficult to see. Changed-case reasoning: A negative local **slope** suggests increasing a locally, not that every positive jump decreases loss.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

A **slope** describes how the loss changes near the current parameter. The symmetric calculation compares equal small moves on either side. Its negative result suggests a small increase may help at zero.

It does not promise every increase helps: jumping to five passes the optimum at two and produces a larger loss. Check the new loss after each proposed step.

Limit: A finite difference approximates **local change**; numerical precision and nonsmooth points need care in more complex objectives.

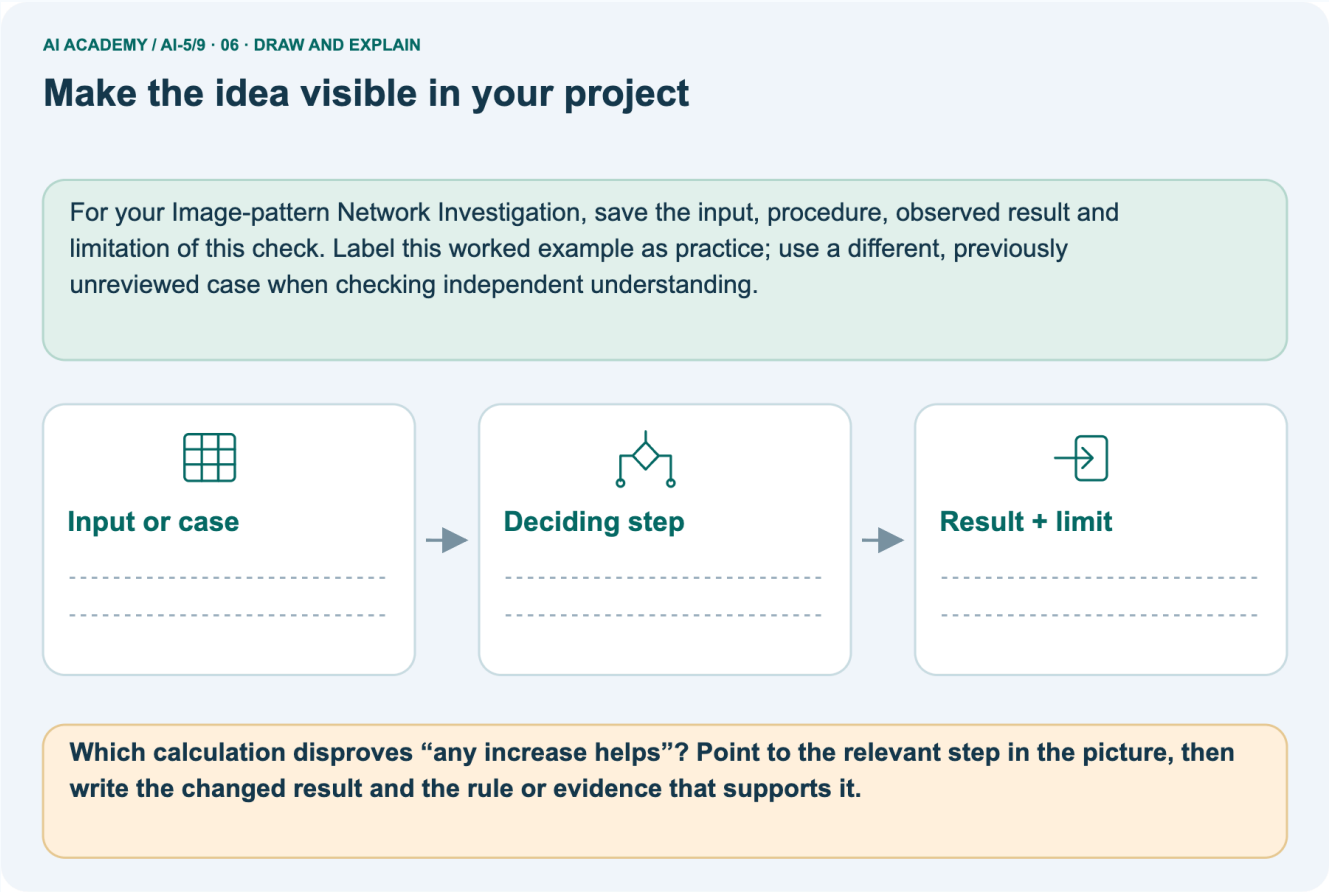
Explain the case, then point to a limit on the claim.

Discuss while showing the picture

Teacher prompt: Explain the case, then point to a limit on the claim. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Keep the stated scope: A finite difference approximates **local change**; numerical precision and nonsmooth points need care in more complex objectives.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For your Image-pattern Network Investigation, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: Which calculation disproves “any increase helps”? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Draw your project trace and justify the deciding step using **evidence**.

Discuss while showing the picture

Teacher prompt: Draw your project trace and justify the deciding step using **evidence**. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Supported practice key: $J(5)=(5-2)^2=9>J(0)=4$. The local direction is useful only with an appropriately sized step and verification. Changed-case check: Loss becomes 9, exceeding the starting 4.

A negative local **slope** suggests increasing a locally, not that every positive jump decreases loss.

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.